

Challenge (Habanero)

Q1. A gamer's high score is 45% of the total points. If the total points are 1200, what is the gamer's high score in points?

- (A) 540
 (B) 600
 (C) 540
 (D) 480
 (E) 720

Q2. A new game has 3.5 GB of free space. If it requires an update of 1.8 GB, what fraction of the free space will be left after the update?

- (A)

rac35

(A)

rac12

(A)

rac47

(A)

rac710

(A)

rac910

Q3. To beat a level, a player must collect at least 75% of the coins. If there are 480 coins in total, how many coins must the player collect?

- (A) 300
 (B) 320
 (C) 360
 (D) 380
 (E) 400

Q4. What is the decimal equivalent of $\frac{7}{8}$?

- (A) 0.78
 (B) 0.80
 (C) 0.85
 (D) 0.875
 (E) 0.90

Q5. A game developer earns 2.5% of the game's total revenue. If the game earns

1,200,000

, how much does the developer earn?

- (A)

24,000

- (B)

30,000

- (C)

36,000

- (D)

45,000

- (E)

60,000

Q6. A player's score is 25% more than their opponent's. If the opponent's score is 120 points, what is the player's score?

- (A) 140

- (B) 150

- (C) 160

- (D) 180

- (E) 200

Q7. A game requires 0.5 GB of RAM to run. What fraction of 2 GB is this?

- (A)

rac12

- (A)

rac13

- (A)

rac14

- (A)

rac15

- (A)

rac16

Q8. A gamer has 18.5 hours of gaming time left. How many hours will they have left after playing for 2.5 hours?

- (A) 16

- (B) 16.5

- (C) 18

- (D) 20

- (E) 21

Open Ended Questions (FRQ)

Q9. A new update increases the game size by 15%. If the current size is 2.8 GB, how much larger will the update be than the current size in GB? Show your work and express your answer as a decimal.

Q10. If a gamer earns 250 points per hour and they have 3.75 hours to play, how many points will they earn in total? Express the answer as a product of a whole number and a decimal.

Chef's Tips

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- Tip 1: Ensure students use exact decimal equivalents and fractions for calculations.
- Tip 2: Remind students to consider significant figures in their calculations.
- Tip 3: Encourage students to use visual aids like number lines for free response questions.